

Linear equations

linear equations: A first order DE of the form $a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$ is said to be linear equation in the variable y

Standard form is require, you need to get the equation into standard form before you start solving. Standard form looks like this:

$$\frac{dy}{dx} + P(x)y = f(x)$$

Method of solution

note: $\frac{d}{dx} [M(x)y] = \underbrace{M \frac{dy}{dx} + \frac{dM}{dx} y}_{\text{product rule}} \Rightarrow M \frac{dy}{dx} + M' y$

→ this will always be **integrating factor $\cdot y$** , there will always be a y

integrating factor

$$M(x) = e^{\int P(x) dx}$$

• the $P(x)$ will always be in front of y in standard form

↑
integrating
factor

how to go about solving

- 1) get the equation in standard form
- 2) identify $P(x)$, find the integrating factor using $M(x) = e^{\int P(x) dx}$
- 3) multiply both sides of the standard form equation by the integrating factor
- 4) then apply:

$$\frac{d}{dx} [M(x)y] = e^{\int P(x) dx} F(x)$$

4) integrate both sides of the equation and solve for y

Practice Problems

$$1) \frac{dy}{dx} + y = e^{3x}$$

• this is already in standard form

$$P(x) = 1, \quad e^{\int 1 dx} \Rightarrow e^x$$

integrating factor

• take e^x and multiply both sides of the equation

$$e^x \frac{dy}{dx} + e^x y = e^{3x} e^x$$

$$\frac{d}{dx} [e^x y] = e^{3x} e^x$$

integrating factor y

• now take the integral in both sides

$$\int \frac{d}{dx} [e^x y] dx = \int e^{4x} dx$$

$$e^x y = \frac{1}{4} e^{4x} + c$$

$$y = \frac{1}{4} e^{3x} + \frac{c}{e^x}$$

$$2) x \frac{dy}{dx} + 2y = 3x$$

begin by getting the equation into standard form

$$\frac{dy}{dx} + \frac{2}{x} y = 3$$

• all we did was factor out x and divide to the other side

$$P(x) = \frac{2}{x}, \quad e^{\int \frac{2}{x} dx} \Rightarrow e^{2 \ln x} \Rightarrow e^{\ln x^2} = x^2 \leftarrow \text{integrating factor}$$

multiply both sides of the standard form equation by the integrating factor

$$x^2 \frac{dy}{dx} + \frac{2}{x} y \cdot x^2 = \frac{3}{x} \cdot x^2$$

$$\underbrace{\hspace{10em}}_{\frac{d}{dx} [x^2 y]} = 3x$$

integrate both side

$$\int \frac{d}{dx} [x^2 y] dx = \int 3x dx$$

$$x^2 y = \frac{3x^2}{2} + C$$

$$\boxed{y = \frac{3}{2} + \frac{C}{x^2}}$$

$$3) \quad \frac{dy}{dx} = 2x - 3y, \quad y(0) = \frac{1}{3}$$

get the equation into standard form

$$\frac{dy}{dx} + 3y = 2x$$

$$P(x) = 3, \quad e^{\int 3 dx} \Rightarrow e^{3x} \leftarrow \text{integrating factor}$$

multiply both side of the standard equation by the integrating factor

$$e^{3x} \frac{dy}{dx} + 3y e^{3x} = 2x e^{3x}$$

$$\underbrace{\hspace{10em}}_{\frac{d}{dx} [e^{3x} y]} = 2x e^{3x}$$

↪ integration by parts

Find the u-value for integration by parts using **liPet**

$$u = 2x$$

$$du = 2dx$$

$$dv = e^{3x}$$

$$v = \int e^u \frac{du}{3}$$

$$v = \frac{1}{3} e^{3x}$$

$$u = 3x$$

$$du = 3dx$$

$$\frac{du}{3} = dx$$

$$uv - \int v du$$

$$(2x) \left(\frac{1}{3} e^{3x} \right) - \int \frac{1}{3} e^{3x} dx$$

$$\frac{2}{3} x e^{3x} - \frac{1}{9} e^{3x}$$

$$e^{3x} y = \frac{2}{3} x e^{3x} - \frac{1}{9} e^{3x} + C$$

$$y = \frac{2}{3} x - \frac{1}{9} + \frac{C}{e^{3x}}$$

$$y(0) = \frac{1}{3}$$

↗ ↖

$$\frac{1}{3} = \frac{2}{3} \cancel{x(0)} - \frac{1}{9} + \frac{C}{e^{3(0)}} \Rightarrow \frac{1}{3} + \frac{1}{9} = C$$

$$C = \frac{4}{9}$$

$$y = \frac{2}{3} x - \frac{1}{9} + \frac{4}{9e^{3x}}$$

$$4) \quad x \frac{dy}{dx} - y = x^2 \sin(x)$$

get this equation into standard form

$$\frac{dy}{dx} - \frac{y}{x} = x \sin(x)$$

• we factored out a x
and we divided two
the other side

$$P(x) = -\frac{1}{x}, \quad e^{\int -1/x \, dx} \Rightarrow e^{-\ln|x|} \Rightarrow e^{-\ln|x|^{-1}} \Rightarrow \frac{1}{x}$$

↑
integrating
factor

multiply both sides of the standard form equation by the integrating factor

$$\frac{1}{x} \frac{dy}{dx} - \frac{1}{x} y = \frac{1}{x} x \sin(x)$$

$$\underbrace{\frac{d}{dx} \left[\frac{1}{x} y \right]} = \sin(x)$$

$$\int \frac{d}{dx} \left[\frac{1}{x} y \right] dx = \int \sin(x) dx$$

$$\frac{1}{x} y = -\cos(x) + c$$

$$y = -x \cos(x) + xc$$

transient terms

these are terms that eventually tend to go to zero as x gets large

$$\text{So, } \lim_{x \rightarrow \infty} \Rightarrow x \Rightarrow 0$$

• when you are being asked for transient terms you apply the limit to each of the solutions terms and see which of them go to zero those are transient terms

You usually find transient terms after you have solved the equation:

ex:

$$y = x^2 + \frac{3}{x} + \frac{c}{x^2} \quad \text{Find the transient terms}$$

take the limit of every term

$$\lim_{x \rightarrow \infty} x^2 = \infty$$

Not a transient term because it goes to infinite not zero

$$\lim_{x \rightarrow \infty} \frac{3}{x} = 0$$

transient term because it goes to 0 as $x \rightarrow \infty$

$$\lim_{x \rightarrow \infty} \frac{c}{x^2} = 0$$

transient term because it goes to 0 as $x \rightarrow \infty$

so the transient terms are: $\frac{3}{x}, \frac{c}{x^2}$

$$5) \quad y dx - 3(x+y^5) dy = 0$$

get this into standard form

$$1 - \frac{3}{y}(x+y^5) \frac{dy}{dx} = 0$$

• all we did was factor $y dx$ and divide to the other side

$$1 - \frac{3}{y}x + 3y^4 \frac{dy}{dx} = 0$$

• we distributed $\frac{3}{y}$

$$1 - \frac{3}{y}x = -3y^4 \frac{dy}{dx}$$

$$\frac{dx}{dy} - \frac{3}{y}x = -3y^4$$

• notice the only way to write it in standard form is by having dx/dy

$$P(x) = -\frac{3}{y}, \quad e^{-\int 3/y \, dy} \Rightarrow e^{-3 \ln y} \Rightarrow e^{-\cancel{3} \ln y^{-3}} \Rightarrow \frac{1}{y^3}$$

integrating factor $\rightarrow$

instead of solving for "y" we are solving for "x" remember if you want to know what you are solving for you look for numerator of $\frac{dx}{dy}$ which is "x"

multiply the integrating factor to both sides of the standard form equation

$$\underbrace{\frac{1}{y^3} \frac{dx}{dy} - \frac{3}{y} x \cdot \frac{1}{y^3}}_{\frac{d}{dy} \left[\frac{1}{y^3} x \right]} = 3y^4 \cdot \frac{1}{y^3}$$

$$\frac{d}{dy} \left[\frac{1}{y^3} x \right] = 3y$$

integrate both sides

$$\int \frac{d}{dy} \left[\frac{1}{y^3} x \right] dy = \int 3y \, dy$$

$$\frac{1}{y^3} x = \frac{3}{2} y^2 + c$$

$$x = \frac{3}{2} y^5 + y^3 c$$